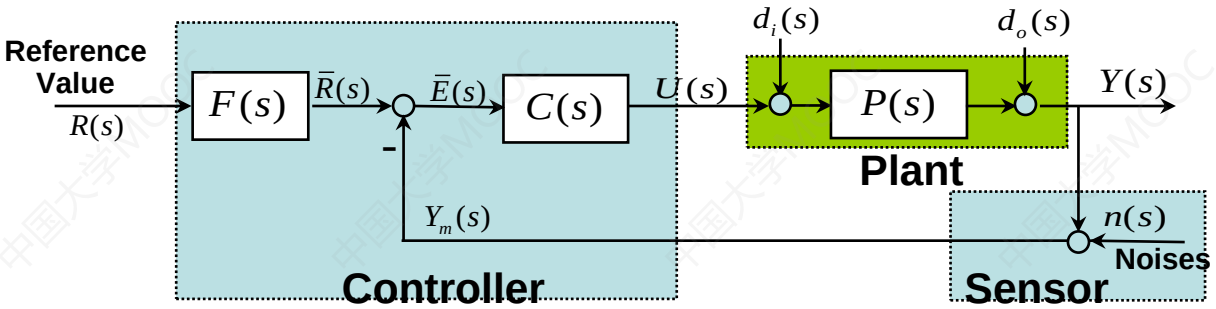
Transient Performance Specifications

(瞬态性能指定)

Tracking Error of Two-DOF Controller



- **CL Response in s-Domain** (s域闭环响应)

$$Y(s) = \underbrace{T(s)F(s)}_{T_r(s)} R(s) + S_i(s)d_i(s) + S(s)d_o(s) - T(s)n(s)$$

$L(s) = P(s)C(s)$, open-loop TF

$$S(s) = \frac{1}{1+L(s)} = \frac{D_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$T(s) = \frac{L(s)}{1+L(s)} = \frac{N_p(s)N_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

$$S_i(s) = \frac{P(s)}{1+L(s)} = \frac{N_p(s)D_c(s)}{D_p(s)D_c(s)+N_p(s)N_c(s)}$$

- **Tracking Error in s-Domain** (s域跟踪误差)

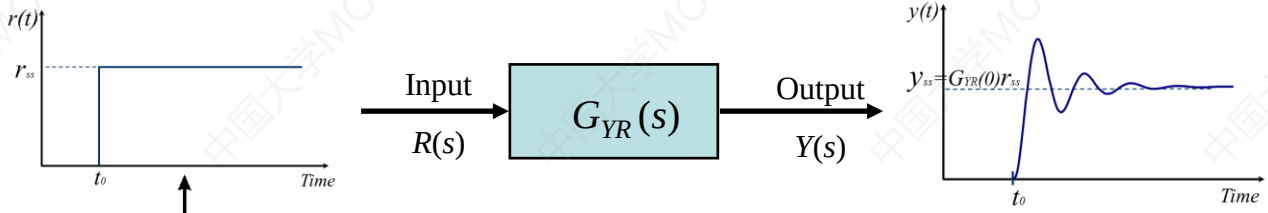
$$E(s) = R(s) - Y(s)$$

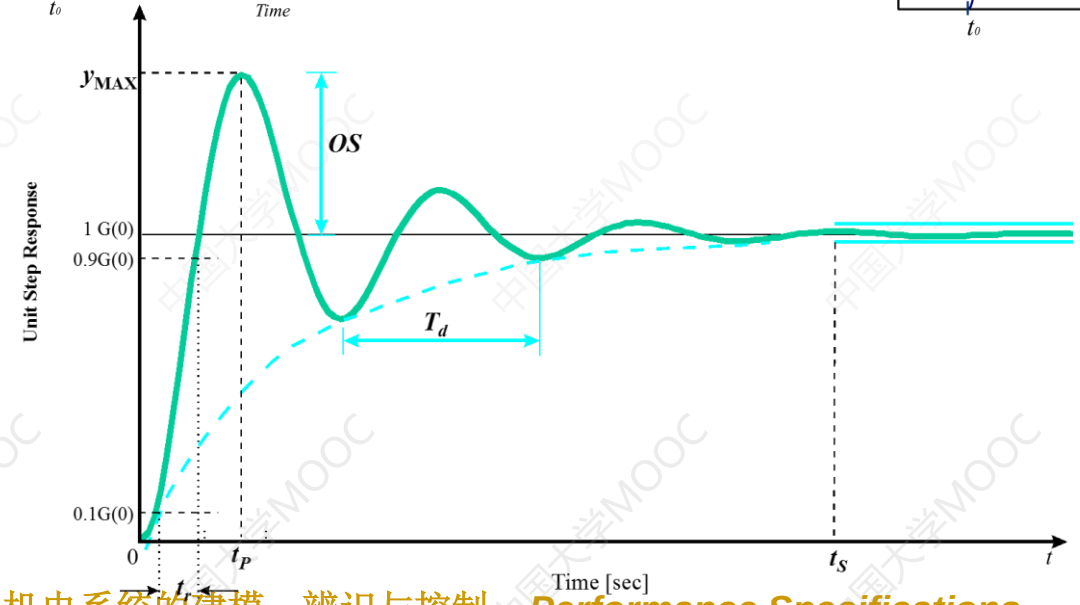
$$= \underbrace{\left(1-T_r(s)\right)R(s)}_{E_r(s)} \underbrace{-S_i(s)d_i(s)}_{E_{di}(s)} \underbrace{-S(s)d_o(s)}_{E_{do}(s)} \underbrace{+T(s)n(s)}_{E_n(s)}$$

error due to reference input | error due to input disturbance | error due to output disturbance | error due to measurement noise

Transient Performance Specifications

- **Traditionally Specified Based on Unit Step Response**
  (基于阶跃响应的传统性能指定)





Transient Performance Specifications

- **Transient Performance (Transient Response)** (瞬态性能)

Transient performance is usually specified using the unit step response of the system (瞬态性能通常通过系统单位阶跃响应来指定)

– **(i) Settling Time (t_S)** (调整时间)

Specifies the time required for the response to reach and stay within a specific percent of the final (steady-state) value (指定了达到并呆在稳态值附近允许误差范围内所需的响应时间)

For a 2nd order system without zero (对于2阶无零点系统):

$$p_{1,2} = \underbrace{-\zeta\omega_n}_{\sigma} \pm j\underbrace{\omega_n\sqrt{(\zeta^2-1)}}_{\omega_d} \Longrightarrow \begin{cases} \dfrac{4}{\zeta\omega_n} & \text{for 2\% bound} \\ \dfrac{5}{\zeta\omega_n} & \text{for 1\% bound} \end{cases} \leq \text{Desired Settling Time } (T_{sd})$$

– **(ii)% Overshoot** (%**OS**) (2nd order system without zero) (超调量)

$$\%OS = 100\% \times e^{-\pi\frac{\zeta}{\sqrt{1-\zeta^2}}} \leq X\%$$

Transient Performance Specifications and CLTF Poles

Assuming that the unit step response of a feedback control system can be approximated by a second-order system without zeros, the requirement on the transient response of the feedback control system can be met by constraining the locations of the poles of the closed-loop transfer function (CLTF) at certain regions on the complex plane (假设反馈控制系统的单位阶跃响应能用二阶无零点系统来近似,为满足反馈控制系统一定的瞬态响应要求，可通过使闭环传递函数的极点位置约束在复平面的某一范围域内来实现):

Dominant CL Poles (主导闭环极点):

$$s_{1,2} = -\zeta\omega_n \pm j\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\omega_d} = -\sigma \pm j\omega_d \,.$$

σ

Settling Time (2%) (调整时间): → ***Puts constraint on the real part of the dominating closed-loop poles*** (设置主导闭环极点的实部约束条件)

$$t_S(2\%) = \frac{4}{\zeta\omega_n} = \frac{4}{\sigma} \le T_{sd} \quad \Longrightarrow \quad \sigma \ge \frac{4}{T_{sd}}$$

%OS (超调量): → ***Puts constraint on the damping ratio ζ or the angle φ of the dominating closed-loop poles*** (设置主导闭环极点的阻尼比或与实轴夹角的约束条件)

$$\%OS = 100\% \times e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \le X\% \quad \Longrightarrow \quad \zeta \ge \sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2 + \left[\ln\left(\frac{X}{100}\right)\right]^2}}$$

Performance Specification → CL Pole Locations

- **Transient Performance Specifications and CLTF Pole Locations**

(瞬态性能指定和闭环传递函数极点位置)

Transient performance specifications can be interpreted as constraints on the locations of the poles of the closed-loop transfer function. Let a pair of closed-loop poles be represented as **(瞬态性能指定可理解为对闭环传递函数极点位置的约束):**

$$p_{1,2} = -\sigma \pm j\omega_d$$

Transient Performance Specifications **(瞬态性能指定):**

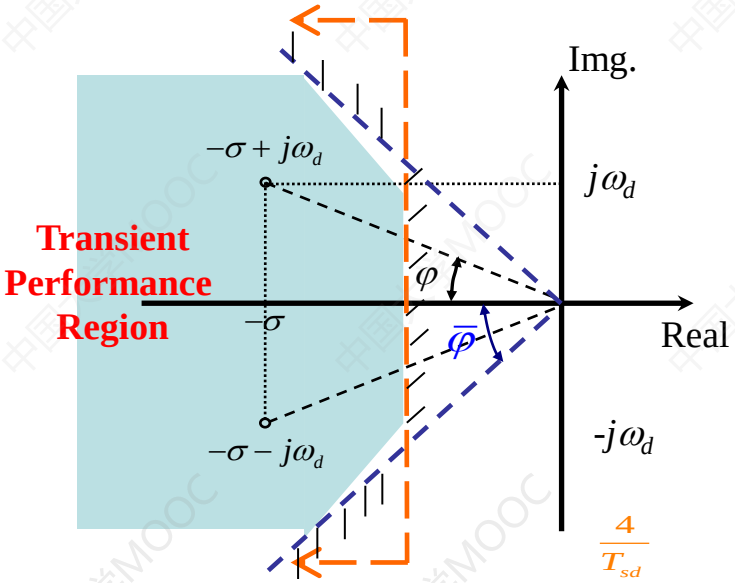
—*Settling Time (2 %)* **(调整时间)** $\leq T_{sd}$

$$t_S(2\%) = \frac{4}{\sigma} \leq T_{sd} \quad \Rightarrow \sigma \geq \frac{4}{T_{sd}}$$

— *%OS* **(超调量)** $\leq X\%$

$$\%\mathrm{OS} = 100\% \times e^{-\pi \cdot \frac{\sigma}{\omega_d}} \leq X\%$$

$$\Rightarrow \tan(\varphi) \leq -\frac{\pi}{\ln(X\%)} \Leftrightarrow \varphi \leq \underbrace{\arctan\left(-\frac{\pi}{\ln(X\%)}\right)}_{\bar{\varphi}}$$



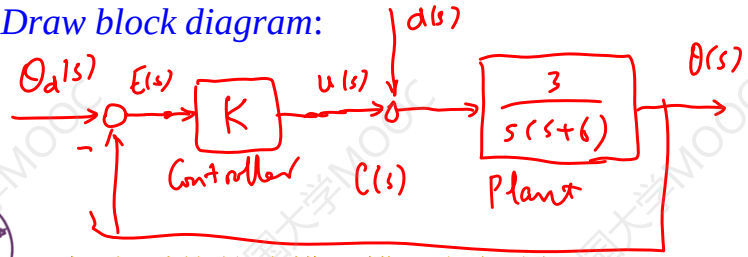
Example

A DC motor driven positioning system can be modeled by a second order transfer function **(直流电机驱动的定位系统用以下二阶传递函数来建模):**

$$G(s)=\frac{3}{s(s+6)}$$

A proportional feedback control is proposed and the proportional feedback gain is chosen to be 16/3. Find the closed-loop transfer function, as well as the 2% settling time and the percent overshoot of the closed loop system when given a step input **(采用比例反馈控制器，比例反馈增益为16/3,找出闭环传递函数和系统在阶跃响应下的2%调整时间和超调量).**

Draw block diagram:



Find closed-loop transfer function:

$$\theta(s)=G_{YR}(s)\theta_d(s)+G_{YD}d(s)$$

$$\text{where}: G_{YR}(s)=\frac{C(s)G(s)}{1+C(s)G(s)}=\frac{K\dfrac{3}{s(s+6)}}{1+K\dfrac{3}{s(s+6)}}=\frac{3K}{s(s+6)+3K}$$

$$G_{YR}(s)=\frac{3K}{s^2+\underbrace{6}_{2\zeta\omega_n}s+\underbrace{3K}_{\omega_n^2}}$$

$$\Rightarrow\begin{cases}\omega_n=\sqrt{3K}\\ \zeta=\dfrac{6}{2\omega_n}=\dfrac{3}{4}\end{cases}$$

$$G_{YR}(0)=1$$

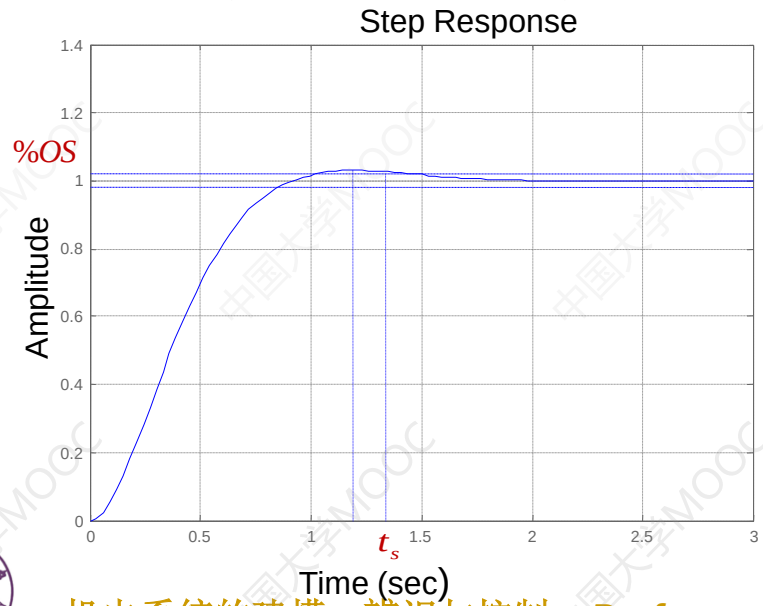
$$G_{YD}(s)=\frac{G(s)}{1+G(s)C(s)}=\frac{\dfrac{3}{s(s+6)}}{1+K\dfrac{3}{s(s+6)}}=\frac{3}{s^2+6s+16}$$

Example

Find closed-loop poles:

$$s^2 + 6s + 16 = 0$$

$$\Rightarrow p_{1,2} = \underbrace{-3}_{\sigma} \pm \underbrace{\sqrt{7}}_{\omega_d} j \quad \Rightarrow \quad \begin{aligned} \sigma &= \zeta\omega_n \\ \omega_d &= \sqrt{1-\zeta^2}\,\omega_n \end{aligned}$$



2% settling time:

$$t_s = \frac{4}{\sigma} = \frac{4}{3} = 1.33\text{sec}$$

%OS:

$$\%OS = 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$= 100 \times e^{-\frac{\pi\frac{3}{4}}{\sqrt{1-(\frac{3}{4})^2}}}$$

$$= 2.84\%$$

Example

A DC motor driven positioning system can be modeled by a second order transfer function(直流电机驱动的定位系统用以下二阶传递函数来建模):

$$G(s)=\frac{3}{s(s+6)}$$

A proportional feedback control is proposed. It is desired that (提出比例反馈控制，期望达到):

- *(P1) for a unit step response, the steady state position should be within 2% of the desired position* (单位阶跃响应的稳态位置在期望位置2%的范围内),
- *(P2) the 2% settling time should be less than 2 sec, and* (2%调整时间小于2秒)
- *(P3) the percent overshoot should be less than 10%* (超调量小于10%).

Find (找出)

(1) *the condition on the proportional gain such that the steady state performance is satisfied* (满足稳态性能要求的比例增益的条件);

(2) *the allowable region in the complex plane for the closed-loop poles* (闭环极点在复平面上的允许范围).

Find closed-loop transfer function:

$$G_{YR}(s)=\frac{C(s)G(s)}{1+C(s)G(s)}=\frac{3K}{s(s+6)+3K}$$

$$\Rightarrow p_{1,2}=-3\pm\sqrt{9-3K}$$

Write down the performance specifications:

To meet,

$$(P1):\quad \left|\frac{\theta_{ss}-\theta_d}{\theta_d}\right|\le 2\%=0.02,\quad \left|\frac{G_{YR}(0)\theta_d-\theta_d}{\theta_d}\right|\le 0.02$$

$$\Rightarrow\quad 0.98\le G_{YR}(0)\le 1.2$$

$$(P2):\quad t_s\le 2\sec\Leftrightarrow\sigma\ge\frac{4}{t_s}=2$$

$$(P3):\quad OS\%\le 10\%\Rightarrow 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\le 10\%$$

$$\Rightarrow\quad \zeta\ge\sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2+\left[\ln\left(\frac{X}{100}\right)\right]^2}}=0.5912$$

Example

Steady state performance constraint

(稳态性能约束):

$$0.98 \le G_{YR}(0) = 1 \le 1.02$$

Transient performance constraint

（瞬态性能约束):

2% Settling Time (调整时间)

$$\sigma \ge 2$$

Percent Overshoot (%OS) (超调量)

$$OS\% \le 10\%$$

$$\varphi \le \bar{\varphi} = \arctan\left(-\frac{\pi}{\ln(X\%)}\right)$$

$$= \tan^{-1}\left(-\frac{\pi}{\ln(0.1)}\right) = 53.76^{o}$$

Closed-Loop Poles ($K>3$):

$$s^2 + 6s + 3K = 0 \quad \Rightarrow \quad s_{1,2} = \underbrace{-3}_{\sigma} \pm j\underbrace{\sqrt{3K-9}}_{\omega_d}$$

$$\therefore \quad \frac{\sqrt{3K-9}}{3} = \tan(\varphi) < \tan(\bar{\varphi}) \quad \Rightarrow \quad K < 8.6$$

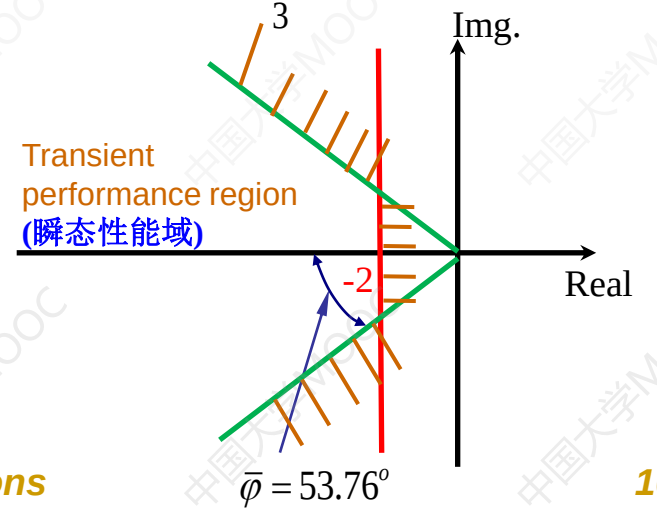
Closed-Loop Poles ($K<3$):

$$s^2 + 6s + 3K = 0 \quad \Rightarrow \quad s_{1,2} = -3 \pm \sqrt{9-3K}$$

$$\therefore \quad -3 + \sqrt{9-3K} < -2 \quad \Rightarrow \quad K > \frac{8}{3}$$

For CL Poles to be in Transient Performance Region (闭环极点在瞬态性能域内):

$$\Rightarrow \frac{8}{3} < K < 8.6$$



PERFORMANCE SPECIFICATIONS

- **Transient Performance Specifications (瞬态性能指定)**
  - Unit Step Response (单位阶跃响应)
  - Second-order Equivalence Approximation (dominant poles concept) (二阶等效近似(主导极点概念))